INCH-POUND

MIL-S-14560F (AL)
1 March 1994
SUPERSEDING
MIL-S-14560E (AL)
26 September 1985

MILITARY SPECIFICATION

SPRAY OUTFIT, PAINT; FOR USE WITH STRIPPABLE COMPOUND

This specification is approved for use within the US Army Armament Research, Development and Engineering Center, Department of the Army and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of spray outfit capable of agitating, force feeding (see 3.2), and spraying strippable compound.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-P-704	- Pails, Metal: (Shipping, Steel, 1 through 12 Gallons).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-P, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4940

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116	- Preservation, Methods of
MIL-C-16555	- Coating Compound, Strippable, Sprayable.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-101	- Federal Test Method Standard, Test Procedures for Packaging Materials

MILITARY

MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of US Military Property
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing, With Appropriate Test Methods
MIL-STD-1190	- Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service, Bldg 4D, 700 Robbins Ave., Philadelphia, PA 19111.)

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

Code of Federal Regulation 29, Chapter XVII	- Occupational Safety and Health Administration Act (OSHA)

(Application for copies should be addressed to the Superintendent of Document, U.S. Government Printing Office, Washington, DC 20402)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME - Boiler and Pressure Vessel Codes.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951 - Standard Practice for Commercial Packaging.
ASTM E380 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 Design. The spray outfit shall be designed to agitate strippable compound to a homogenized spraying consistency, to either pump or pressure feed (manufacturer's option), to regulate the air pressure, and to spray the strippable compound through an external atomizing-type spray gun. The design shall be such that parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement without modification, distortion, or damage to the spray outfit or any component.

3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the spray outfit. In this specification, all measurements, dimensions, sizes and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380.

Source: https://assist.dla.mil -- Downloaded: 202640-18T03:26Z
Check the source to verify that this is the current version before use.

* 3.2.2 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances that will provide for the interchangeability of respective parts between end items of the same model. All replacement parts shall be interchangeable with their respective original replacement parts, and fit/function with mating components.

* 3.2.3 Threads. All machined threads shall conform to FED-STD-H28 and the applicable Detailed Standard referenced therein.

3.2.4 Safety and health requirements. The spray outfit shall comply with the general safety and health requirements promulgated under Code of Federal Regulation 29, Chapter XVII - OSHA that are applicable to the outfit itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.6).

* 3.3 Materials. Materials not specifically designated herein or in the contract shall be of a quality commensurate with best commercial practices, shall be suitable for the intended purpose in the design and fabrication of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would affect the performance, maintainability, reliability, durability, or longevity of the individual components or the end item equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction. The spray outfit shall be furnished complete so that, when installed, it can be used for the operations specified herein.

3.4.1 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

Source: https://assist.dla.mil -- Downloaded: 202640-18T03:26Z
Check the source to verify that this is the current version before use.

* 3.4.4 Lubricating systems. All bearings (except sealed for life, permanently lubricated type), matching gears, and all other moving parts installed in the end item equipment shall be provided means to assure adequate lubrication. Circulating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. All oil holes, grease fittings, and filler caps shall be so located as to be readily accessible. The supplier shall include and install all lubricants, greases and other lubricating compounds. Vent openings in the lubrication systems shall be temporarily capped or plugged to prevent loss of lubricants during shipment and handling. Tags, warning that "ALL TEMPORARY CAPS OR PLUGS MUST BE REMOVED BEFORE START-UP OR USE OF THE ITEM", shall be attached to all temporary caps or plugs intended to be vented to the atmosphere during operation. These warnings shall also be included in the instruction manual. In the event a lubricant reservoir cannot be covered and sealed, it shall be appropriately preserved and the lubricant shall be provided in a sealed container and shipped with each piece of end item equipment. A warning tag shall be placed in a conspicuous place near the filling area, warning the user to "INSTALL THE FURNISHED LUBRICANT BEFORE OPERATION OF THE EQUIPMENT". The operator's manual shall also reflect this above warning and procedure.

3.4.5 Surfaces. Surfaces of castings, forgings, stampings, and parts shall be clean and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials.

3.4.6 Protective finishes. Exposed metal (see 6.7.1) and working surfaces, except when the parts involved are inherently corrosion resistant, shall be protected with an electrochemical or chemical conversion finish.

3.5 Performance and product characteristics. The spray outfit, when attached to a filtered air supply from 80 to 120 pounds per square inch gage (psig) pressure at a volume of 30 cubic feet per minute (cfm), shall be capable of pumping or pressure feeding and spraying strippable compound of MIL-C-16555 at room temperature. The spray outfit shall provide continuous agitation of the compound.

3.5.1 Continuous duty. The spray outfit and all components shall be capable of withstanding eight hours of continuous operation each day for ten days, with strippable compound conforming to MIL-C-16555 being agitated, pumped or pressure fed during that entire period of operation. The spray outfit shall be able to withstand the continuous operation without damage, malfunction, leakage, distortion, or impairment of operation.

3.6 Components.

3.6.1 Cover assembly.

3.6.1.1 Pressure feed unit (when provided). The cover assembly shall be a metal stamping or forging, shall be self-centering, and shall have a sealing gasket providing an air and liquid tight seal with the pressure tank specified

Source: https://assist.dla.mil -- Downloaded: 202640-18T03:26Z
Check the source to verify that this is the current version before use.

in 3.6.8. The cover assembly shall not be deformed when the sealing force is applied by the tank fastening devices specified in 3.6.8. The cover assembly shall conform to the ASME Boiler and Pressure Vessel Code (see 2.2) for the pressure specified in 3.6.8.

* 3.6.1.2 Pump feed unit (when provided). The Sheet-Metal used for the cover assembly shall be not less then Standard Gage No. 20 in accordance with Manufacturer's Standard Gage for Steel Sheet. The cover assembly shall completely shield and fit over the open top of a five gallon container conforming to type II, class 2 or 3, of PPP-P-704 or a container housing, if furnished. If a container housing is furnished, it shall be fabricated of Sheet-Metal not less then Standard Gage No. 20 in accordance with Manufacturer's Standard Gage for Steel Sheet and shall be large enough to completely enclose a five gallon container conforming to type II, class 2 or 3, of PPP-P-704. Provision shall be made for securing the cover assembly to the five gallon container or container housing. The cover assembly or container housing shall incorporate two carrying handles.

3.6.2 Pump assembly (for pump feed unit only). The pump assembly shall be mounted on the cover assembly in 3.6.1.2 and shall be an air motor driven rotary-gear or reciprocating-type unit. The air motor shall operate satisfactorily on the air supply specified in 3.5 to drive the pump for delivering the strippable compound to the spray gun. The pump intake shall be located near the bottom of the container so that no more than 2-1/4 inches of strippable compound shall be left in the emptied container when the container is sitting on a level surface.

3.6.3 Agitator. Each spray outfit shall be provided with an air motor-type agitator capable of operating (in conjunction with other pneumatic components) from the air supply specified in 3.5. The agitator shall be designed to bring the contents of the full container to the consistency and viscosity that shall assure satisfactory application of the strippable compound under the conditions specified in 3.5. A manually operated speed adjusting device shall be provided to shut off any or all air to the motor. The agitator drive may be consolidated with the pump motor drive shaft when pumps are used for liquid feed.

3.6.4 Air pressure control. A regulator, or control valve, shall be provided to control the air pressure to the spray gun. The regulator, or control valve, shall be capable of withstanding pressures up to and including 200 psig. Parts exposed to compressed air shall be constructed of corrosion-resistant materials. The pressure regulator shall be capable of being adjusted to provide a minimum air pressure regulator range of 0 to 100 psig.

3.6.5 Liquid pressure control. Means shall be provided to control the liquid pressure to the spray gun within a minimum range of 0 to 100 psig and be capable of withstanding pressures up to, and including, 200 psig.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

3.6.6 Air pressure gage. An air pressure gage shall be provided to indicate the air pressure to the spray gun. The pressure gage shall be a Bourdon tube type. The dial shall be not less than two inches in diameter. Graduations and figures shall be of simple, legible design and on a contrasting background. The gage shall have a minimum pressure scale range of 0 to 100 psig. The scale shall be provided with divisions of not more than five psig, with figure intervals of not more than 20 psig. The error in pressure indication shall not exceed five psig on the middle third of the scale, and ten psig on the upper and lower thirds of the scale.

3.6.6.1 Gage dampeners (for pump feed unit only). Gages shall be equipped with pulsation dampeners to reduce wear on the gage movement, and to preclude the possibility of fluid entry into the Bourdon tube. The dampeners shall in no way detract from the accuracy specified in 3.6.6.

3.6.7 Liquid pressure gage. A gage shall be provided to indicate the liquid pressure to the spray gun. Requirements for the liquid pressure gage shall be the same as for the air pressure gage specified in 3.6.6, except that the liquid pressure gage shall have a minimum pressure scale range of 0 to 160 pounds psig.

3.6.8 Pressure tank (for pressure feed unit only). The pressure tank shall be fabricated of steel, shall have a capacity of not less than five gallons, and shall have a galvanized interior and exterior. The tank shall be of seamless or welded construction, cylindrical in shape, having a full open top and concave bottom with welded bottom chime. The pressure tank shall be fitted with fastening devices to seal and lock the cover assembly in position to withstand, without leakage or damage, a test pressure of not less than 165 psig hydrostatic pressure for a period of five continuous minutes. The tank material outlet shall extend near the tank bottom so that not more than 2-1/4 inches of strippable compound shall remain in the emptied tank when the tank is sitting on a level surface. The pressure tank shall incorporate two carrying handles located near the top edge which are diametrically opposite each other. The pressure tank shall be certified for conformance to the ASME Boiler and Pressure Vessel Codes, with a maximum working pressure of 110 psig. The name-plate required by the ASME code shall be permanently attached to the cover or chime. ASME certification shall be acceptable in lieu of an additional pressure (hydrostatic) test (see 4.6.5).

3.6.9 Base. A base shall be provided with means to mount and secure the spray outfit equipped with strippable compound. The base shall be fitted with at least three swivel wheels or swivel casters, not less than two inches in nominal diameter, so that the spray outfit is maneuverable.

* 3.6.10 Spray gun. One heavy duty pressure feed paint spray gun NSN 4940-00-261-8413 shall be provided as Government-furnished property (see 6.3).

3.6.11 Air and liquid hoses. The following air and liquid hoses shall be provided. Certification of compliance shall be furnished by the manufacturer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

* 3.6.11.1 Air hoses. Two air hoses, each not less than 25 feet in length, shall be provided with each spray outfit. The hoses shall have an oil-resistant tube and cover, shall be fiber or textile reinforced, and shall have a rated working pressure of not less than 200 psig. One hose shall connect the spray gun to the output side of the air pressure regulator or control, and the other hose shall connect the air compressor to the air supply inlet on the cover assembly. The hose from the regulator or control output to the spray gun shall have an inside diameter of not less than 3/8 inch. The hose from the compressor to the cover assembly inlet shall have an inside diameter of not less than 1/2 inch. The air hose couplings shall be swivel, female, straight pipe thread, with a 60 degree included angle taper seat except that the air hose for connecting the compressor to the cover shall have a quick-disconnect coupling. The air hose to the spray gun shall include the necessary adapters to fit the air connection to the spray gun specified in 3.6.10.

* 3.6.11.2 Liquid hose. A liquid hose, not less than 1/2 inch inside diameter and not less than 25 feet in length, shall be provided to connect the pump feed or pressure feed unit to the spray gun. The hose shall be compatible with compounds conforming to MIL-C-16555, and shall be rated for a working pressure of not less than 750 psig. Each end shall have a swivel female coupling with standard straight pipe thread, with a 60 degree included angle taper seat. The liquid hose shall include the necessary adapters to fit the liquid connection of the spray gun specified in 3.6.10.

3.6.12 Valves. Valves shall be constructed to seal and provide positive adjustment and shall be fabricated of noncorrosive materials if exposed to compressed air. Control valves shall be provided at the air operated motors and at the air or liquid system junctions or bypasses on the cover, or at the spray gun. Adjustment ranges shall be provided to insure operation as specified herein.

3.6.12.1 Safety valves. Safety valves shall be provided on pressure tanks to insure that dangerous pressures do not develop. Release pressures for safety valves shall be governed by the ASME Boiler and Pressure Vessel Codes (see 3.6.8).

3.7 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.8 Product identification. The spray outfit shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

3.9 Workmanship. Standards of workmanship shall assure that the spray outfit shall have the stability, strength, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply will all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to a preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the product examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

* 4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (f) below. If a First Article Test (FAT) is not specified, one unit from the first lot shall be subjected to the packaging validation test in 4.6.6 and 4.6.6.1. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Functional test (see 4.6.2).
(c) Accuracy of pressure gages test (see 4.6.3).
(d) Pressure test (see 4.6.5).
1/(e) Packaging validation test (see 4.6.6).
(f) Packaging inspection (see 4.7).

* 4.4.1 Sampling. Sampling for quality conformance inspection shall be performed as listed with all samples selected randomly.

Inspection or test	Lot or Batch Size	Sample Size
(a) Product Examination (4.5)	1-150	13
(b) Functional test (4.6.2)	1-150	13
(c) Accuracy of pressure gages test (4.6.3)	1-150	13
(d) Pressure test (4.6.5)	1-150	13
2/(e) Packaging validation test (4.6.6)		1
(f) Packaging inspection (4.7)	1-150	5

The lots shall not exceed the maximum sizes indicated above. If lot size is less than or equal to sample size, 100 percent inspection is required. Each lot shall be accepted with no defects and rejected if one or more defects are found.

4.5 Product examination. Visually, dimensionally, and manually examine each item to determine conformance with the requirements of 3.2 through 3.4.6 and 3.6.9 through 3.9. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Dimensional examination includes measuring dimensions as specified. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for

Notes:

1. Perform the packaging validation test on one unit only when a FAT has not been conducted.
2. Perform the packaging validation test on one unit from the first production lot only when a FAT has not been conducted.

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:26Z
Check the source to verify that this is the currentversion before use.

acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the Item and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit or function of the end item (see 3.2.2), shall be cause for rejection.

4.6 Tests.

4.6.1 Performance test. (For first article testing.) When stabilized at an ambient air temperature of +72 degrees to +78 degrees F, the spray outfit shall be loaded with five gallons of strippable compound conforming to MIL-C-16555, connected to an air source of 30 cfm at 80 to 120 psig, and operated in accordance with the manufacturer's instructions. The spray gun shall be directed into a clean receptacle and the spray outfit shall be operated until the strippable compound container is emptied. During, and upon conclusion of the test, the spray outfit shall be examined to determine compliance with the following:

	REQUIREMENT	REFERENCE
(a)	Continuous duty test (see 4.6.4)	3.5.1
(b)	Cover assembly for pressure feed unit	3.6.1.1
(c)	Cover assembly for pump feed unit	3.6.1.2
(d)	Pump assembly for pump feed unit	3.6.2
(e)	Agitator	3.6.3
(f)	Air pressure control	3.6.4
(g)	Liquid pressure control	3.6.5
(h)	Air pressure gage	3.6.6
(i)	Liquid pressure gage	3.6.7
(j)	Pressure tank for pressure feed unit	3.6.8
(k)	Base	3.6.9
(l)	Spray Gun	3.6.10
(m)	Air hoses	3.6.11.1
(n)	Liquid hose	3.6.11.2
(o)	Valves	3.6.12

4.6.2 Functional test. (For quality conformance testing.) Spray outfit components shall be tested in accordance with the manufacturer's standard procedures to verify compliance with 4.6.1 (b) through (o). Tests may be performed at the individual component level provided compliance with all performance requirements is verified.

4.6.3 Accuracy of pressure gages test. Accuracy of the air and liquid pressure gages shall be tested at all major graduations throughout the entire range of the scale by comparison with a standard pressure measuring device having an overall accuracy within one percent of the value being measured (see 3.6.6 and 3.6.7).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

4.6.4 Continuous duty test. (For first article testing.) In an ambient air temperature of +72 degrees to +78 degrees F, the spray outfit shall be filled with strippable compound conforming to MIL-C-16555 and connected to an air supply of 30 cfm at 80 to 120 psig. The spray outfit shall be subjected to eight hours of continuous operation each day for ten days (see 3.5.1). When a pump feed unit is furnished, it shall be set up to spray the strippable compound back into the container. Continuous operation shall be considered to be the period of operation, less refill time, with the delay for refilling not to exceed ten minutes. The spray outfit shall agitate, feed, and spray the strippable compound throughout the test with no evidence of damage, malfunction, leakage, or impairment of operation (see 3.5.1).

4.6.5 Pressure test (for pressure feed unit only). This pressure test shall be conducted only when the manufacturer cannot provide formal ASME certification (see 3.6.8).

4.6.5.1 Test procedure. The release valve and other parts subject to damage shall be removed. All openings shall be plugged and the pressure tank, with the cover assembly secured in place, shall be subjected to a hydrostatic pressure of 165 psig for a period of five continuous minutes. During, and upon conclusion of, the test period, the pressure tank shall be examined for evidence of leakage, deformation, or other damage (see 3.6.1.1 and 3.6.8).

* 4.6.6 Packaging validation test. After the Quality Assurance inspection and test has been successfully completed, one shipping container carrying one unit shall be subjected to the edgewise-drop (rotational) test (4.6.6.1) specified in FED-STD-101. The container shall include a unit packaged in accordance with the requirements of Section 5, PACKAGING. If a FAT is required by the contract, the shipping container carrying the paint spray outfit subjected to the continuous duty test (4.6.4) shall be selected to conduct the test. If a FAT is not required by the contract, then a shipping container carrying a paint spray outfit from the first production lot shall be used for the test. The test apparatus used shall be as described in FED-STD-101. A dummy load shall not be used. All tests are to be conducted under ambient conditions. Environmental temperature/humidity tests are not required. After the test has been completed, the container shall be inspected. Defects in accordance with those described in 4.6.6.1 shall be cause for rejection. If there is evidence of possible damage to the tested unit which cannot be visually determined, the paint spray outfit shall then be unpacked and shall again undergo the Quality Assurance inspections and tests of paragraphs 4.5, 4.6.2, and 4.6.3 to verify that the unit has not been damaged. After all the requirements have been met, the unit shall be repackaged before submission to the Government for acceptance inspection of paragraph 4.7. After testing and inspection, any damage to the packaging shall be repaired prior to acceptance of the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

* 4.6.6.1 Edgewise-drop (rotational) test. Test shall be conducted using Method 5008.1 of FED-STD-101. Make 4 drops, one on each edge of the container base, using drop height as listed in Table 1. Criteria for passing/failing of the test is stated in Paragraph 7.1.5 of Method 5008.1.

4.7 Packaging inspection. The items shall be inspected before and after packaging to determine compliance with the preservation, packaging, packing, and marking requirement specified in Section 5.

5. PACKAGING

5.1 Disassembly. When practical, items shall be disassembled to afford protection of components, attachments and accessories against damage, pilferage and to reduce cubage. Disassembled parts shall be clearly and legibly marked as to identity and proper location on the assembled item and matchmarked when necessary to facilitate reassembly. Removed parts and mating parts on the item shall be matchmarked by stenciling on the part or by the use of tags. Tags shall be waterproof and ink used shall be waterproof. Tags shall be firmly secured to each individual part. All fasteners removed during disassembly shall be reinstalled in one of the mating parts. A part shall not be disassembled unless it can be readily assembled in the field without special skills or tools.

5.2 Requirements. Thorough cleaning and drying of all bare surfaces is essential before the application of a preservative. Cleaning, drying and application of preservatives shall progress in an uninterrupted series of operations. When periods of interruption are necessary, temporary protection shall be provided to partially processed items, as required, to insure against contamination. The materials used in cleaning, drying and preservation shall not be injurious to the item being processed.

a. All cleaning shall be in accordance with method C-1 of MIL-P-116, except that no class 1 ozone depleting substances such as 1.1.1 trichloroethane, halons, chlorofluorocarbons, or carbon tetrachloride are authorized for use.

b. All preservation shall be in accordance with MIL-P-116. Petroleum (P-type) preservative P-6 is not authorized for use. Petroleum (P-type) preservatives shall not be applied to non-metallic parts, corrosion resistant metal or metal that is painted or treated to resist corrosion.

5.3 Preservation. Preservation shall be level A, C or Commercial as specified.

5.3.1 Level A.

5.3.1.1 Cleaning and drying. Component parts, disassembled parts, attachments, accessories, assembly hardware and small miscellaneous parts shall be cleaned, dried, preserved, packaged and packed like the basic unit. Drying shall be accomplished by any applicable means stated in MIL-P-116.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

5.3.1.2 Preservation. Immediately after cleaning and drying, surfaces that are subject to corrosion or deterioration shall be treated with a preservative. Excessive use of preservative compounds shall be avoided.

5.3.1.3 Technical data. Technical data furnished shall be placed in a waterproof, greaseproof bag, sealed and placed in the unit container housing the spray outfit.

5.3.1.4 Electrical components. Electrical equipment shall be clean and dry. Preservatives shall be applied as required to prevent corrosion to moving or other metal parts, but shall not come into contact with electrical connections or windings. Openings shall be sealed with cap-plugs or with waterproof, pressure sensitive tape to provide protection against entry of any foreign materials. A tag shall be placed in a conspicuous place warning "ALL TEMPORARY CAPS, PLUGS OR TAPE MUST BE REMOVED BEFORE START-UP OR USE OF THIS MACHINE". The above warning shall also be included in the instruction manual.

5.3.1.5 Cables, wires and hoses. Exposed ends of wires, hoses, cables, plug openings, sockets, connector plugs, terminals and openings into switches shall be sealed with waterproof tape or cap-plugs. All cables, hoses and wires whether attached or unattached shall be coiled to a minimum safe diameter to prevent kinking and twisting and secured in three equidistant places with cotton twine or tape to prevent uncoiling.

5.3.1.6 Instruments. All instruments, (gages, dial indicators, meters, switches) and all surfaces of the machine where abrasive action could interfere with the form, fit or function of the machine shall be covered with a 1 inch thick uncompressed cushioning material and secured in place with pressure sensitive tape. Cushioning materials and tape used shall be water resistant, non-dusting and non-corrosive.

5.3.1.7 Drive belts. Drive belts shall either be removed or released from tension. Cog belts which do not rely solely on tension do not have to be released from tension but, care must be taken to assure that the belts do not deteriorate or stretch while the machine is in storage. Metal pulleys shall be coated with a rust inhibiting primer.

5.3.2 Level C. Level C preservation shall be in accordance with MIL-STD-1190.

5.3.2.1 Commercial. Commercial preservation shall be in accordance with ASTM D3951.

5.4 Packing. Packing shall be level A, B, C or Commercial as specified.

5.4.1 General. For the intermediate and shipping containers the arrangement of contents, cushioning, anchoring, blocking and bracing shall be in accordance with MIL-STD-1186 except delete "MIL-P-19644" in paragraph 5.3.3.

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:26Z
Check the source to verify that this is the current version before use.

* 5.4.2 Level A.

* 5.4.2.1 General. Before being packaged in the shipping container cushioning shall be applied to prevent marring, puncturing, damage or any abrasive action to the finished surfaces or protruding parts. All cushioning material shall be secured by taping or tying to prevent any possibility of shifting or becoming detached during transit and storage. Cushioning material shall be at least 3/4 inch thick, fabricated in the form of pads or blankets, and made of fiberboard. Disassembled parts, accessories and attachments shall be consolidated, cushioned to prevent damage and packed in fiberboard boxes conforming to PPP-B-636 class WWVR. Federal Specification identified as MMM-A-250 in Federal Specification PPP-B-636 is not authorized for use. The boxes shall be sized to fit the parts being packed. Size and weight limitations of the container specification shall apply.

5.4.2.2 Shipping container. The unit and its associated items shall be packed in a shipping container conforming to PPP-B-601 or PPP-B-621, overseas type. Selection of the container shall be governed by the size and weight limitations of the container specification. Each shipping container shall contain only one unit and all associated items.

5.4.3 Level B. Unless otherwise specified, Level B packing will be the same as level A, except the shipping container shall be of the domestic type.

5.4.4 Level C. Level C packing shall be in accordance with MIL-STD-1190.

5.4.5 Commercial. Commercial packing shall be in accordance with ASTM D3951.

* 5.6 Marking. All markings, whether military or commercial, for all interior and exterior containers shall be in accordance with the requirements as specified in MIL-STD-129. Bar Code marking are required and shall be in accordance with MIL-STD-129.

5.6 Workmanship. The quality of work shall be such that when proper procedure is followed, materials and equipment being processed will be provided the maximum protection against corrosion and deterioration consistent with the shipment hazards known or anticipated, and suitable for storage.

5.7 Rough handling. Rough handling shall be in accordance with Method 5008.1 of FED-STD-101.

6. NOTES

6.1 Intended use. The spray outfit covered by this specification is intended for use in agitating, pump or pressure feeding, and spraying strippable compound.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) First article when required (see 3.1).
(c) Exceptions and additional safety/health requirements when required (see 3.2.4 and 6.6).
(d) Painting and finishing if different (see 3.7).
(e) Product identification if different (see 3.8).
(f) First article inspection if different (see 4.3).
(g) Quality conformance inspection if different (see 4.4).
(h) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

* 6.3 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.6.10.

6.4 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

* 6.5 Subject term (key word) listing.

Agitator
Air pressure control
Pressure feed
Pump feed
Spray Outfit

6.6 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.7 Definitions

6.7.1 Exposed metal. Exposed metal is defined as metal exposed to atmospheric conditions, compressed air, solvents, and the material being sprayed (see 3.4.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

6.8 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodian:

Army -AL

Preparing Activity:

Army - AL

Project No. 4940-A616

Source: https://assist.dla.mil -- Downloaded: 202640-18T03:26Z
Check the source to verify that this is the currentversion before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-14560F(AL)	2. DOCUMENT DATE (YYMMDD) 1 MARCH 1994

3. DOCUMENT TITLE
Spray Outfit, Paint; for use with Strippable Compound

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial) Cmdr, USA Armament Res, Dev. & Engr Ctr	b. ORGANIZATION	
c. ADDRESS (Include Zip Code) ATTN: SMCAR-EST-P Rock Island, IL 61299-7300	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. NAME Cmdr USA Armament Res, Dev. & Engr Ctr	b. TELEPHONE (Include Area Code) (1) Commercial (319) 782-8147	(2) DSN 793-8147
c. ADDRESS (Include Zip Code) ATTN: SMCAR-EST-P Rock Island, IL 61299-7300	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 — Previous editions are obsolete. — 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:26Z
Check the source to verify that this is the current version before use.